

NON-PARAMETRIC MODELS

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What is Non-Parametric?

A linear parametric model takes the form

$$Y = \beta X + \epsilon$$

A non-linear parametric model takes the form

$$Y = \beta X_1^{\beta_1} X_2^{\beta_2} + \epsilon$$

A non-parametric model takes the form

$$Y = m(X) + \epsilon$$

where m is a function of a conditional expectation $m(X) = E[Y|X]$, provided the exogeneity assumption of regressors $E[\epsilon_t|X = x] = 0$. We focus on estimation at a single point x which is in the interior of the support of X .

The goal is to estimate $\hat{m}(X)$ without taking a strong stand on the functional form of $m(X)$.

Review of Estimation Methods

Review of methods that aim to estimate:

- 1 Binned Means Estimator → Tuning parameter: bandwidth h
- 2 Kernel Regression → Tuning parameter: bandwidth h
- 3 Series Regression → Tuning parameter: number of series p
- 4 Local Polynomial regression → Tuning parameter: h and p

Binned Means Estimator

Fix the point x and consider estimation of $m(x)$. This is the mean of p for random pairs (p, X) such that $X = x$. This binned means estimator can be written as

$$\hat{m}(x) = \frac{\sum_{t=1}^T \mathbb{1}[|X_t - x| \leq h] Y_t}{\sum_{t=1}^T \mathbb{1}[|X_t - x| \leq h]}$$

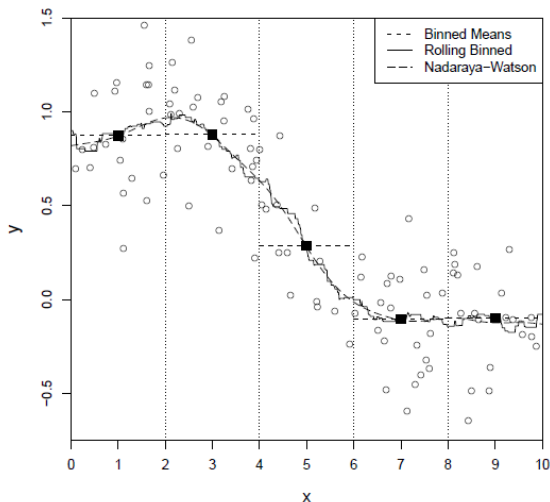
This is an step function estimator of the regression function $m(x)$.

The binned estimator $\hat{m}(x)$ is the step function which is constant within each bin and equals the binned mean.

A fine grid of values of x and thereby obtain a smoother estimate. This estimator is called Rolling Binned Means.

Binned Means Estimator

Figure 1: Binned Means Estimator



Note: The observations are displayed as the open circles. The estimator of $m(x)$ at $x = 1$ with $h = 1$ is the average of the p_t for the observations such that X_t falls in the interval $[0 \leq X_t \leq 2]$. Source: [Hansen, \(2021\)](#).

Kernel Regression

One deficiency with the binned mean estimator is that it is a step function in x even when evaluated on a fine grid.

The source of the discontinuity is that the weights are discontinuous indicator functions.

If instead the weights are continuous functions then $\hat{m}(x)$ will also be continuous in x

The kernel regression estimator can be written as

$$\hat{m}(x) = \frac{\sum_{t=1}^T \mathcal{K}\left(\frac{X_t - x}{h}\right) Y_t}{\mathcal{K}\left(\frac{X_t - x}{h}\right)}$$

The estimator is also known as the Nadaraya-Watson estimator, or the local constant estimator.

Kernel Regression

Common Normalized Second-Order Kernels

- Rectangular: $\mathcal{K}(u) = \begin{cases} \frac{1}{2\sqrt{3}} & \text{if } |u| < 3 \\ 0 & \text{otherwise} \end{cases}$
- Gaussian: $\mathcal{K}(u) = \frac{1}{2\pi} \exp\left(-\frac{u^2}{2}\right)$
- Epanechnikov: $\mathcal{K}(u) = \begin{cases} \frac{3}{4\sqrt{5}} \left(1 - \frac{u^2}{5}\right) & \text{if } |u| < \sqrt{5} \\ 0 & \text{otherwise} \end{cases}$
- Triangular: $\mathcal{K}(u) = \begin{cases} \frac{1}{\sqrt{6}} \left(1 - \frac{|u|}{\sqrt{6}}\right) & \text{if } |u| < \sqrt{6} \\ 0 & \text{otherwise} \end{cases}$

Series Regression

A series regression model is a sequence $S = 1, 2, \dots$ Of approximating models $m_S(x)$ with S parameters. we focus on linear series models, and in particular polynomials and splines.

Linear series regression models take the form

$$Y = X_S' \beta_S + \epsilon$$

where $X_S = X_S(X)$ is a vector of regressors and β_S is a coefficient vector

The series approximation to $m(x)$ is

$$m_S(x) = X_S'(x) \beta_S$$

The coefficient is estimated by least squares $\hat{\beta}_S = (X_S' X_S)^{-1} X_S' Y$. Then,

$$\hat{m}_S(x) = X_S'(x) \hat{\beta}_S$$

Series Regression-Polynomial Regression

The prototypical series regression model form $m(x)$ is a p^{th} order polynomial

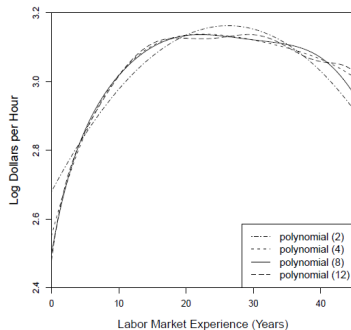
$$m_S(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_p x^p.$$

where $X_S = (1 \ x \ \dots \ x^p)'$ and $\beta = (\beta_0, \beta_1, \dots, \beta_p)'$.

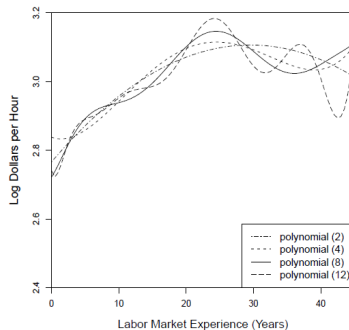
The implied polynomial regression model for the random pair (Y, X) is $Y = X_S' \beta_S + \epsilon$ with $X_S = (1 \ X \ \dots \ X^p)'$.

Series Regression-Polynomial Regression

Figure 2: Polynomial Estimates



(a) White Women



(b) Black Women

Note: Polynomial Estimates of Experience Profile, College-Educated Women. Source: [Hansen \(2021\)](#).

Series Regression-Splines

- A spline is a piecewise polynomial
- Typically the order of the polynomial is pre-selected to be linear, quadratic, or cubic
- The flexibility of the model is determined by the number of polynomial segments
- The join points between the segments are called **knots**

A linear spline with one knot τ is

$$m_S(x) = \beta_0 + \beta_1 x + \beta_2(x - \tau)\mathbb{1}\{x \geq \tau\}$$

A linear spline with two knots $\tau_1 < \tau_2$ is

$$m_S(x) = \beta_0 + \beta_1 x + \beta_2(x - \tau_1)\mathbb{1}\{x \geq \tau_1\} + \beta_3(x - \tau_2)\mathbb{1}\{x \geq \tau_2\}$$

Series Regression-Splines

A quadratic spline with one knot is

$$m_S(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 (x - \tau)^2 \mathbb{1}\{x \geq \tau\}$$

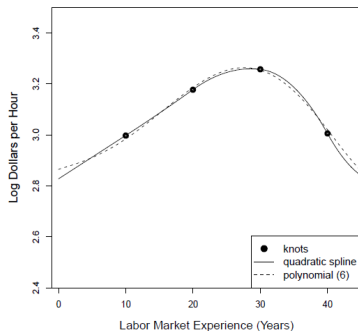
The implied spline regression model for the random pair (Y, X) is $Y = X_S' \beta_S + \epsilon$ where

$$X_S = \begin{pmatrix} 1 \\ X \\ X^2 \\ (X - \tau)^2 \mathbb{1}\{X \geq \tau\} \end{pmatrix}$$

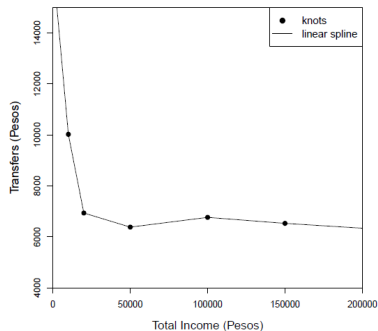
The model can be generalized to a p_{th} -order spline with N knots. In practice a spline will depend critically on the choice of the knots.

Series Regression-Splines

Figure 3: Spline Regression Estimates



(a) Experience Profile



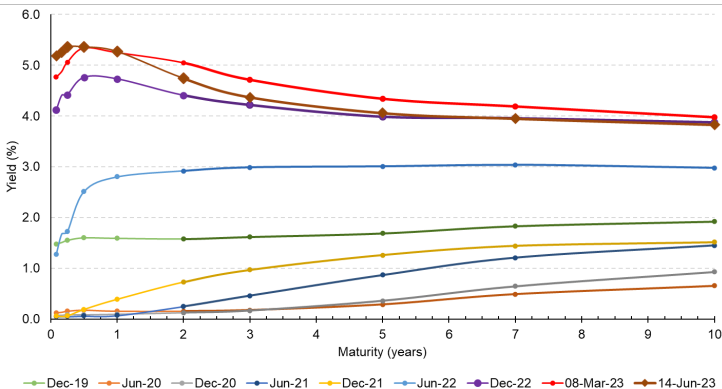
(b) Effect of Income on Transfers

Note: Polynomial Estimates of Experience Profile, College-Educated Women. Source: [Hansen \(2021\)](#).

Yield Curves

The yield curve is a graph which depicts how the yields on debt instruments – such as bonds – vary as a function of their years remaining to maturity.

Figure 4: U.S. Treasury Yield Curves



Note: Each curve starts at the 1-month treasury yield. Curve at the end of the month, mid yield to maturity. On March 08, 2023, Silicon Valley Bank released its Mid-Quarter update and disclosed a loss of \$1.8 billion USD in the sale of securities, including Treasury and mortgage bonds. Source: Bloomberg.

Some Definitions

We start with the simple case of zero bonds, which pay a fixed amount at some specified date in the future.

- The spot rate is the yield y_t on an t -period zero bond with payoff M and price p

$$p = \frac{M}{(1 + y_t)^t}$$

- The discount factor d_t is the standardized price for unit payoff in the t -period future, and satisfies

$$d_t = \frac{1}{(1 + y_t)^t}$$

Some Definitions

- The forward rate f_t is the rate paid on a one-period investment arranged today and made at time t in the future maturing at time $t + 1$

$$1 + f_t = \frac{d_t}{d_{t+1}}$$

- The yield to maturity y on a coupon bond paying c in each period and 1 at the end of t periods is implied by the relation

$$p = \frac{c}{(1+y)} + \frac{c}{(1+y)^2} + \dots + \frac{1}{(1+y)^t}$$

The par bond yield curve is the graph of the yield to maturity of coupon bonds that sell at par

Some Definitions

The choice of period matters in the above formulae. It is convenient to work with infinitesimal time periods (continuous compounding)

- The discount function at time τ is the price of one dollar received at point τ in the future:

$$d(\tau) = \exp(-\tau y(\tau))$$

where $y(\tau)$ is the yield (the rate of interest on a zero-coupon payment at maturity τ)

- The yield curve is related to the forward rate by

$$y(\tau) = \frac{1}{\tau} \int_0^{\tau} f(s) ds$$

where $f(s)$ is the forward short term rate applicable at time s .

Estimation of the Yield Curve from Coupon Bonds

- In practice, we do not observe a full set of zero coupon bonds, one for each date, but instead we have a sample of bonds many of which are coupon bonds
- From this sample we want to extract the zero coupon yield curve
- In practice, we have data on prices p , payments b , and payment times τ
- We want to estimate the functions d , y , and f

Estimation of the Yield Curve from Coupon Bonds

The statistical model is

$$p_i = \sum_{j=1}^{T_i} b_{ij} d(\tau_{ij}) + \epsilon_i$$

- p_i is the price of bond i at the observation date
- $b_{ij} = b_{ij}(\tau_{ij})$ is the payment to be received by the holder of bond i at time τ_{ij} in the future, where $j = 1, \dots, T_i$
- ϵ_i is a random error term that soaks up small pricing errors, liquidity effects, maturity effects, etc.
- The quantity of interest is the discount function $d(\cdot)$

Yield Curves-Nelson Siegel parametric functional form

- Nelson and Siegel (1987) propose a functional form to 'shape' the yield curve
- 'Just' 4 parameters
- Its structure is sufficiently flexible to capture most yield curves
- But the functional is bound to have just "one hump"
- Svensson (1994) extends the idea to capture more complex curvature structures
- 6 parameters: Over parametrization issues?
- Allows for 2 humps
- The function may have several local maximums

Yield Curves-Nelson Siegel parametric functional form

Nelson and Siegel (1987) specify a parametric model for the forward curve

$$f(t) = \beta_0 + (\beta_1 + \beta_2(t/\tau_0)) \exp(-t/\tau_0)$$

where $\theta(\beta_0, \beta_1, \beta_2, \tau_0)$ are unknown parameters. This implies that the discount function is

$$d_\theta(t) = \exp(-t(\beta_0 + (\beta_1 + \beta_2(t/\tau_0))(1 - \exp(-t/\tau_0))\frac{\tau_0}{t} - \beta_2(-t/\tau_0)))$$

This functional form allows a combination of shapes for the long end and the short end of the yield curve. Estimation of θ is by nonlinear least squares

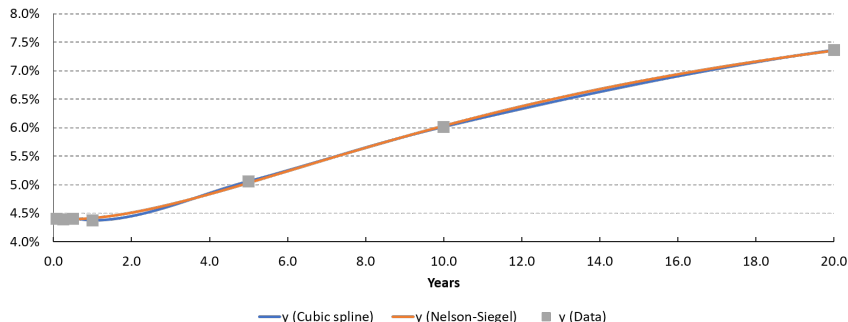
$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \sum_{i=1}^n \left(p_i - \sum_{j=1}^{T_i} b_{ij} d(\tau_{ij}) \right)^2$$

Yield Curves-Splines

- A more parsimonious method to infer the form of the yield curve is to assume that there is no functional form. Then, use an interpolation method
- (Cubic) Splines are sufficiently flexible to capture the curvature of yield curves
- The fixed points are the data. The knots.
- Then, the goal is to use a piecewise (third degree) polynomial to smoothly joint such points
- The US Treasury's yield curve is currently derived using a quasi-cubic Hermite spline function

Yield Curves – Nelson-Siegel vs (Cubic) Splines

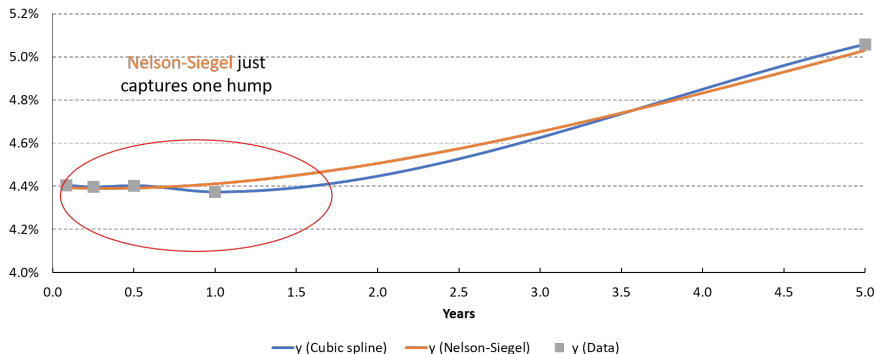
Figure 5: Yield Curves – Nelson-Siegel vs (Cubic) Splines



Source: Course CEMLA: Financial Econometrics (2022).

Yield Curves – Nelson-Siegel vs (Cubic) Splines

Figure 6: Yield Curves – Nelson-Siegel vs (Cubic) Splines



Source: Course CEMLA: Financial Econometrics (2022).

Kernel Densities

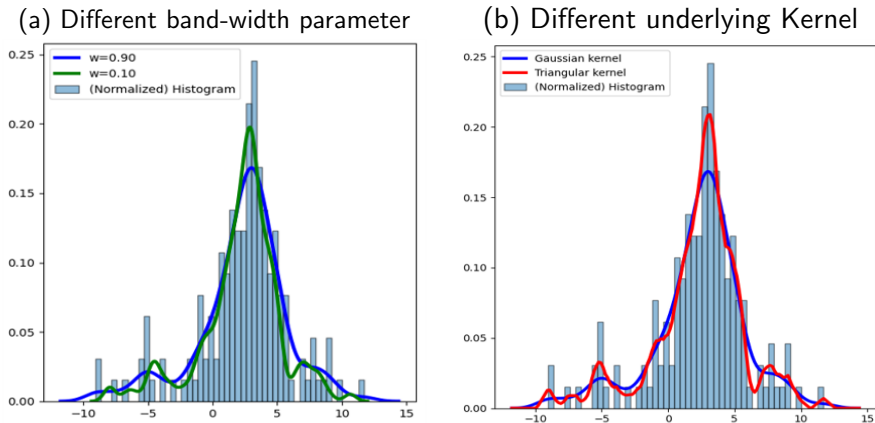
- These are nonparametric methods to estimate probability density functions $f(x)$
- In simplistic terms: to find a smooth curve to fit the histogram of some i.i.d. data $(X_1, X_2, \dots, X_T)$

$$\hat{f}_h(x) = \frac{1}{Th} \sum_{t=1}^T \mathcal{K}\left(\frac{X_t - x}{h}\right)$$

- The Kernel: a non-negative real-valued integrable function $\mathcal{K}$
- $\int_{-\infty}^{\infty} \mathcal{K}(u) du = 1$
- $\mathcal{K}(-u) = \mathcal{K}(u)$
- $h > 0$ is a smoothing parameter called the bandwidth

Kernel Densities-Growth GDP

Figure 7: Mexico GDP y/y Growth: 1981:Q1-2019:Q4



Source: Course CEMLA: Financial Econometrics (2022).

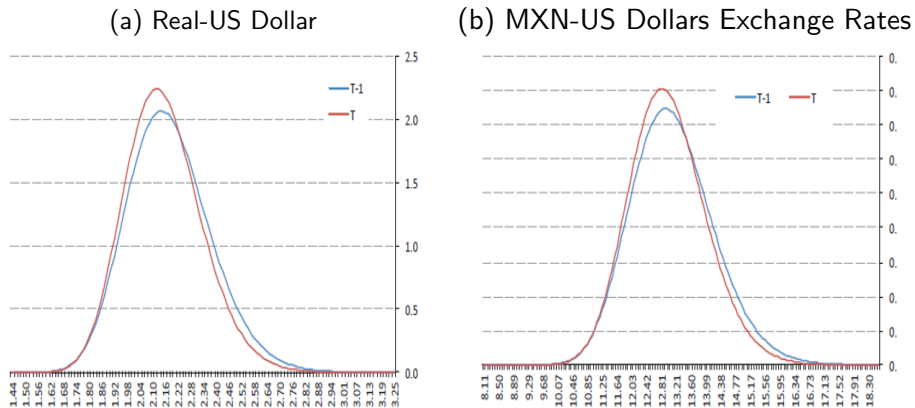
Kernel Densities-Risk-Neutral Densities and FX

- Monetary authorities usually state that they intervene in FX markets not to defend a certain level, but to control the volatility of the market
- Is there a way to measure the volatility of the FX market?
- Using nonparametric distributions, we could also assess if central bank interventions affect other parameters in the distribution
- that the distributions from option prices are Risk-neutral distributions, which provide an estimate of the probability that market participants attach to future price developments
- In the case of exchange rates, they are an important tool in assessing the likelihood of sharp movements - a key risk for financial stability - and the evolution of this likelihood over time

Kernel Densities-Risk-Neutral Densities and FX

The densities are from the Fed's Announcement on the Dollar Swap Lines.
 $T - 1$ indicates the Risk-Neutral Density a Day Before the Announcement.
 T Indicates the Risk-Neutral Density On the Day of the Announcement.

Figure 8: Risk-Neutral Densities

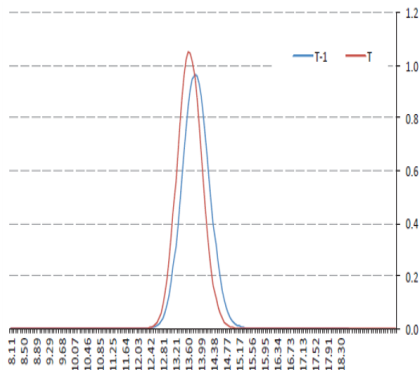


Kernel Densities-Risk-Neutral Densities and FX

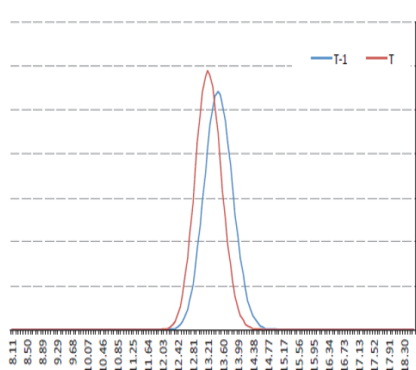
Left Side: Banxico announces its intention to use part of the resources provided by the Dollar Swap Line. Right Side: Banxico implements the auctions using part of the resources associated to the Dollar Swap Line.

Figure 9: Risk-Neutral Densities

(a) MXN-US Dollar Exchange Rate



(b) MXN-US Dollars Exchange Rates



References

- CEMLA (2022, 2021). *Course on Financial Econometrics*.
- Hansen, B.E. (2022b). *Econometrics*. Princeton University Press.
- Linton, O. (2019). *Financial Econometrics Models and Methods*. Cambridge University Press.